

Original article

Randomised clinical trial: Efficacy and safety of Qing-Chang-Hua-Shi granules in a multicenter, randomized, and double-blind clinical trial of patients with moderately active ulcerative colitis

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ABSTRACT

Qing-Chang-Hua-Shi (QCHS) is a Chinese herbal formula, which is composed of 11 herbs. Studies have also shown that QCHS granules can alleviate colitis in animal models by preventing inflammatory responses and suppressing apoptosis through the MEK/ERK signaling pathway. To determine the efficacy and safety of QCHS granules in patients with moderately active UC. We performed a multicenter, randomized, placebo-controlled, double-blind study of patients with moderately active UC who did not respond to 4 weeks of mesalazine therapy at the maximum dose. Patients were randomly assigned to groups and administered QCHS granules (125 g/day, n = 59) or an identical placebo, which was similar to the QCHS granules in color and taste (125 g/day, n = 60), with continued 5-ASA 4 g/d therapy for 12 weeks. The primary outcome was the rate of clinical response and clinical remission at week 12. The secondary outcomes were health-related quality of life, endoscopic response rate, and mucosal healing rate. Any changes in mucus/bloody stool and diarrhea were recorded. Out of the 119 enrolled patients at 10 different centers in China, 102 patients completed the trial. Clinical remission and clinical response were seen in 31.48% and 92.59% of QCHS-treated patients, and 12.50% and 72.92% of placebo-treated patients, respectively. There was a significant difference between the two treatment groups. More patients receiving QCHS granules vs. placebo achieved remission of mucus/bloody stool (70.37% vs. 47.92%, $P = 0.0361$). Adverse event rates were similar (QCHS granules 38.33%; placebo 25.42%). In conclusion, QCHS granules were superior to the placebo in introducing clinical remission and mucosal healing, as well as in relieving mucus/blood stool in patients with moderately active and 5-ASA-refractory UC.

1. Introduction

Ulcerative colitis (UC), a type of inflammatory bowel disease (IBD), is a chronic and nonspecific inflammatory disorder of the mucosa and submucosa of the colon [1]. Patients with UC experience abdominal

pain, diarrhea, and mucus/bloody stool, all of which influence the quality of life [2–4]. According to recent studies, the prevalence of UC in developed countries has reached a peak, while the incidence of this disease is still increasing in developing countries, such as China and other Asian regions [5].

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Although substantial research has been performed over the last few decades, the underlying mechanism of UC, which may correlate with genetic [6,7], environmental [8,9] and immunological factors [10], is still unknown. The mainstay of therapy for the induction of remission in patients with mild to moderate active UC is 5-aminosalicylic acid (5-ASA); however, not all patients respond to 5-ASA [1,11]. Patients with 5-ASA-refractory UC often switch to corticosteroids [12], probiotics [13], immunological therapy [14], and fecal microbiota transplantation [15] (FMT). However, increasing studies have reported the ineffective and adverse events (AEs) related to these treatments, such as disturbed mucosal regeneration, infection, and cushingoid features [16].

In Asian countries, especially in China, Korea, and Japan, herbal medicines have been widely used for approximately two thousand years to treat and manage UC-like symptoms, such as diarrhea, abdominal pain, and mucus/bloody stool. Although the chemical and biological medicines are the mainstream drugs for UC treatment worldwide, herbal medicines [17], as a complementary and alternative medicine, provide another choice for patients with UC in the Asian-Pacific region. In order to manage the symptoms of UC, patients in the Asian-Pacific region have been treated with several types of herbal medicines, such as indigo [18], curcumin [19,20], HMPL-004 (extracted from *Andrographis paniculata*) [21,22], and many other formulas [23,24].

Several ancient classic formulas, such as Bai-tou-wong decoction [25] and Ge-gen-qin-lian decoction [26] from “Synopsis of Golden Chamber,” [27] (written by Zhong-jing Zhang, 150–215 B.C.) as well as Shao-yao decoction [28] from “Simple questions about Qi diseases” [29] (written by Wansu Liu, 1110–1200 B.C.), have been characterized as potential medicine for UC-like symptoms [30]. Qing-Chang-Hua-Shi (QCHS) is derived from the Bai-tou-wong and Shao-yao decoction with modifications by our group. QCHS is composed of *Rhizoma coptidis* (Huanglian), *Radix scutellariae* (Huangqin), *Patrinia mandshurica* (Baijiangcao), *Radix angelicae sinensis* (Danggui), *Radix paeoniae alba* (Baishao), *Radix sanguisorbae* (Diyu), *Radix arnebiae* (Zicao), *Radix et rhizome rubiae* (Qiancao), *Radix angelicae dahuricae* (Baizhi), *Radix Aucklandiae* (Muxiang), and *Radix et rhizome glycyrrhizae* (Gancao) at a ratio of 6:10:15:10:20:12:6:10:10:20:6. An animal study showed its efficacy in alleviating colitis by inhibiting inflammation and protecting intestinal epithelial cells [31]. However, QCHS, as a traditional Chinese medicine, is an empirical medicine, and to-date, no clinical trial has reported the efficacy of QCHS.

The objective of this study was to examine the efficacy and safety of oral administration of QCHS granules in the treatment of UC through a placebo-controlled randomized trial in patients with baseline 5-ASA usage.

2. Methods

2.1. Study design

This was a prospective, multicenter, randomized, double-blind, placebo-controlled, superiority clinical trial, which conformed to the Consolidated Standards of Reporting Trials (CONSORT) 2010 statement guidelines [32] and the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) 2013 Statement [33]. Patients at 10 different centers in China took part in this trial. The trial protocol, included in the [Supplementary Appendix](#), was registered (No. ChiCTR-IOR-14005554) and approved by all ten centers. Ethical approval was obtained from the Ethics Committee of the Affiliated Hospital of Nanjing University of Chinese Medicine (Nanjing, China) and the other nine centers. Before randomization, all participants were informed that they would receive QCHS granules or placebo. The study began in July 2014 and lasted in April 2019. Researchers and patients were blinded from the beginning of the trial. All interested outcomes, including patient-reported outcomes, were collected before (0 week) and after intervention (2, 4, 6, 8, 10, and 12 weeks).

2.2. Participants

Patients were enrolled if they met the following criteria: 1) age 18–65 years; 2) a diagnosis of moderately active UC based on diagnostic criteria; 3) a Mayo score [34,35] of 6–10 (Table 1); 4) took 5-ASA for more than 4 weeks; 5) didn't received the corticosteroids and/or immunosuppressant. Patients were excluded for having at least one of the following conditions: 1) other intestinal diseases (e.g., bacterial dysentery, intestinal tuberculosis, and Crohn's disease); 2) serious intestinal complications (e.g., intestinal perforation, intestinal obstruction, toxic megacolon, colorectal cancer); 3) significant cardiac disease, hepatic disease, pulmonary disease, renal disease, and other serious diseases; 4) severe physical disability (e.g., blindness, deafness, mute, intellectual disability, mental disability, and physical disability); 5) a history of alcohol or drug abuse; and 6) participated in other clinical trials. All the participants provided written informed consent before randomization.

2.3. Randomization and masking

Each eligible patient was randomly allocated into the QCHS granule group (mesalazine + QCHS granules) or control group (mesalazine + placebo) with an allocation ratio of 1:1. Stratified block randomization (by site) was performed to ensure equal group sizes with permuted blocks (block size of 6). The randomization allocation sequence generated by SAS 9.4 (SAS Institute Inc., Cary, NC, USA) was converted into unique serial numbers for each enrolled subject at the subcenters. QCHS granules and placebo granules were provided to each subcenter in advance, encoded, and labeled with the above serial number. To ensure blinding, the QCHS granules and placebo granules were identical in all aspects (e.g., appearance, size, color, smell, taste, containers, and dose). Participant and Principal Investigators were blinded to group allocation throughout the research. The randomization procedure was conducted by an independent research assistant who was not involved in clinical observation or assessment. Under the supervision of the Data Safety and Monitoring Board (DSMB), the randomization code was only broken due to adverse events and statistical analysis. After all study data had been sent to the medical company (Famous Medicine (Jiangsu, China)) for analysis, the data code break for statistical analysis was opened (October 19, 2018).

2.4. Interventions

All participants were given basic treatment with mesalazine sustained-release granules (4 g/d; produced by IPSEN Pharmaceutical Factory, Shanghai, China). The 5-ASA doses were not increased and suppositories, or enema were not used during this subject. Participants in the QCHS granule group received QCHS granules (125 g, twice a day; provided by Jiangyin Tianjiang Pharmaceutical St, Jiangsu, China) or placebo (125 g, twice a day, provided by the same company) orally for 12 weeks. The QCHS granules consisted of 11 herbal products: *Rhizoma coptidis* (Huang Lian, 6 g), *Scutellaria baicalensis* (Huang Qin, 10 g), *Rhizoma sinensis* (Bai Jiang Cao, 15 g), *Radix angelicae sinensis* (Dang Gui, 10 g), *Radix paeoniae alba* (Bai Shao, 20 g), *Radix sanguisorbae* (Di Yu, 10 g), *Radix arnebiae* (Zi Cao, 10 g), *Radix et rhizome rubiae* (Qian

Table 1
Mayo scores of UC activity.

Mayo index	0	1	2	3
Stool frequency	Normal	1–2/day > normal	3–4/day > normal	5/day > normal
Rectal bleeding	None	Streaks	Obvious	Mostly blood
Mucosa	Normal	Mild friability	Moderate friability	Spontaneous bleeding
Physician's global assessment	Normal	Mild	Moderate	Severe

Cao, 20 g), *Radix angelicae dahuricae* (Bai Zhi, 12 g), *Radix Aucklandiae* (Mu Xiang, 6 g), and *Radix et rhizome glycyrrhizae* (Gan Cao, 6 g). Granule of each herb was extracted in hot water, the aqueous extract was concentrated, dried, and packed in sealed packages and batch number of granules of each herb is 14,1134 (Huang Lian), 1,411,335 (Huang Qin), 1,411,336 (Bai Jiang Cao), 1,411,337 (Dang Gui), 1,411,338 (Bai Shao), 1,411,339 (Di Yu), 1,411,340 (Zi Cao), 1,411,341 (Qian Cao), 1,411,342 (Bai Zhi), 1,411,343 (Mu Xiang), 1,411,344 (Gan Cao). Participants were asked to dissolve the granules by hot water and drink the mixture. The dose for each herb in QCHS was designed on the basis of the Chinese Pharmacopoeia. The placebo was with the same granules consisted of maltodextrin, food pigment, a bitter agent. Participants were asked to dissolve the granules by hot water and drink the mixture. During the research, all participants were only received the research drugs without other permitted concomitant drugs. Every two weeks, patients were seen for follow-up visits, and got the following drugs for next two weeks from pharmacist after reporting the usage of drugs.

2.5. Clinical assessment and trial end points

The primary efficacy endpoint was the rate of clinical response and clinical remission after 12 weeks of treatment (last survey point). The clinical response was defined as a 3-point decrease in the Mayo score and a decrease of at least 30% from baseline, along with a decrease of at least 1 point in the rectal bleeding subscore or an absolute rectal bleeding score of 0–1. Clinical remission was defined as a Mayo score ≤ 2 and no subscores with a value greater than 1. The main secondary endpoint was the rate of mucosal response and mucosal healing after 12 weeks of treatment. Mucosal response was defined as a decrease of at least 1 point in the mucosa subscore. Mucosal healing was defined as an absolute mucosa subscore of ≤ 1 . Colonoscopies (Olympus Medical System, Tokyo, Japan or Fujinon Toshiba ES system, Tokyo, Japan) were used to assess the colonoscopic severities. All the colonoscopists with more than five years work experience were trained before taking part in this study. Concomitant medication use was recorded throughout the study. Safety evaluations, a clear indication of relationship between the QCHS treatment and AEs, were assessed throughout the study, including liver dysfunction, renal dysfunction, electrocardiographic abnormality and other abnormal laboratory test. Serum samples collected at baseline and weeks 2, 4, 6, 8, 10 and 12 were assessed for liver dysfunction, which was defined as cases in which aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (AKP), total bilirubin (TBIL), and γ -glutamyl transpeptidase (GGT) levels exceeded the upper limit of each institution at any visit.

2.6. Statistical analysis

On the basis of a previously published trial [36], we assumed the rate of participants with adequate relief of goble IBD to be 41% after placebo treatment. Assuming a significant level of 0.05, a power of 0.8, a difference of 10% was selected as the smallest difference that would be of clinical significance, we need a total of 108 participants. Considering a dropout rate of 10%, we needed a total sample size of 120.

According to our protocol, all data analyses were conducted based on a pre-established statistical analysis plan. All data analyses were performed by SAS software (v. 9.3; SAS Institute Inc., Cary, NC, USA). To ensure consistency and reliability of the conclusions, both intention-to-treat (ITT) analysis and Per-Protocol (PP) analysis were performed if necessary. All missing data was imputed by multiple imputations. Statistical analysis was performed with frequency, mean, median, standard deviation, lower quartile (P25), upper quartile (P75), minimum, and maximum. Normality of all quantitative (continuous) variables (e.g., demographic characteristics, vital signs, and biochemical parameters) was tested by the Kolmogorov-Smirnov test. For normally distributed data, independent sample *t*-tests and paired sample *t*-tests were

employed to compare parameters at the beginning and end of the study between and within groups, respectively. For abnormally distributed data (e.g., quality of life and inflammation markers), the Mann-Whitney *U* test and Wilcoxon signed-rank test were used. For qualitative (categorical) variables, such as clinical response rate, endoscopic response rate, and mucosal healing rate, a chi-square test or Fisher's exact test was used. In the evaluation of efficacy and analysis of influencing factors, center and disease were used as covariates, and covariance analysis (ANCOVA) was used. Multiple linear regression or logistic regression was used to analyze influencing factors and evaluate the effect of gender and condition on the efficacy of the two groups. For all analyses, $P < 0.05$ was considered statistically significant.

3. Results

3.1. Randomization and clinical characteristics of patients at baseline

From April 2014 to April 2019, 122 patients with UC were screened for eligibility, and patients were enrolled and appropriately randomized. All participants received the treatment that they were assigned and safety assessments. The QCHS group included 60 participants. The placebo group included 59 participants. Patient characteristics at baseline are shown in Table 2, and the distribution of the overall extent of the disease was not significantly different among the two groups ($P = 0.8834$). Comparisons of other factors (e.g., age, gender, race, and history of smoking) were not statistically significant between the groups. The PP population included 102 participants, with 90.0% and 81.3% of patients receiving QCHS granules or placebo, respectively, completing the study (Fig. 1).

3.2. Clinical efficacy

QCHS granules have shown a significant effect on alleviating the symptoms of moderate UC in patients. A significantly greater percentage of patients receiving QCHS granules achieved clinical response after 12 weeks (primary efficacy endpoint) of treatment as compared with the placebo (92.59% vs. 72.92%, $P = 0.0093$, Fig. 2A). Clinical remission, another primary efficacy endpoint, was achieved by a significantly greater percentage of patients receiving QCHS granules as compared with the placebo (31.48% vs. 12.50%, $P = 0.049$, Fig. 2A).

The main secondary endpoints were mucosal response and mucosal healing after 12 weeks of treatment. The rates of mucosal response at week 12 were also significantly higher in patients who received QCHS granules compared with patients who received the placebo (83.33% vs. 58.33%, $P = 0.0088$, Fig. 2B). The percentage of mucosal healing at

Table 2
Baseline characteristics of patients with moderately active ulcerative colitis.

Characteristics	QCHS group (n = 60)	Placebo group (n = 59)	P value
Age (y)	44.27 $\pm$ 11.81	39.97 $\pm$ 11.36	0.2725
Female (%)	28 (46.67)	22 (37.29)	0.1437
Race (%)			1.0000
Han	59 (98.3)	58 (98.3)	
Other	1 (1.7)	1 (1.7)	
History of smoking (%)	12 (20)	9 (15.3)	0.461
Clinical data			
Course (month)	55.97 $\pm$ 54.34	56.56 $\pm$ 52.76	0.952
Clinical type, n (%)			0.0815
Chronic relapsing type	58 (96.67)	53 (89.83)	
First episode type	2 (3.33)	6 (10.17)	
Extent of disease, n (%)			0.8834
Rectum	12 (20.00)	11 (18.64)	
Left-side	27 (45.00)	27 (45.77)	
Extensive	21 (35.00)	21 (35.59)	
Parenteral performance, n (%)	2 (3.33)	4 (6.78)	0.3838
Mayo score, n (%)	8.04 $\pm$ 0.93	7.85 $\pm$ 0.94	0.2819

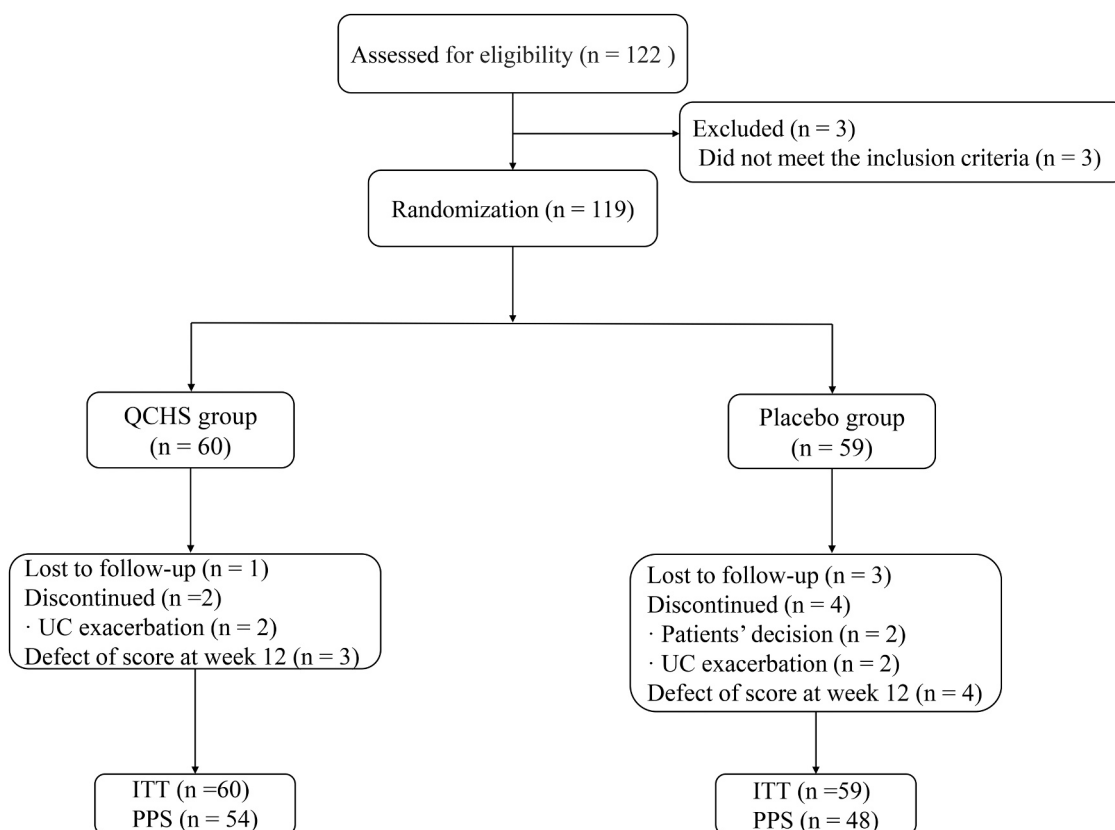


Fig. 1. Study flowchart. QCHS, Qing-Chang-Hua-Shi. UC, ulcerative colitis. ITT, full analysis set. PPS, per protocol set.

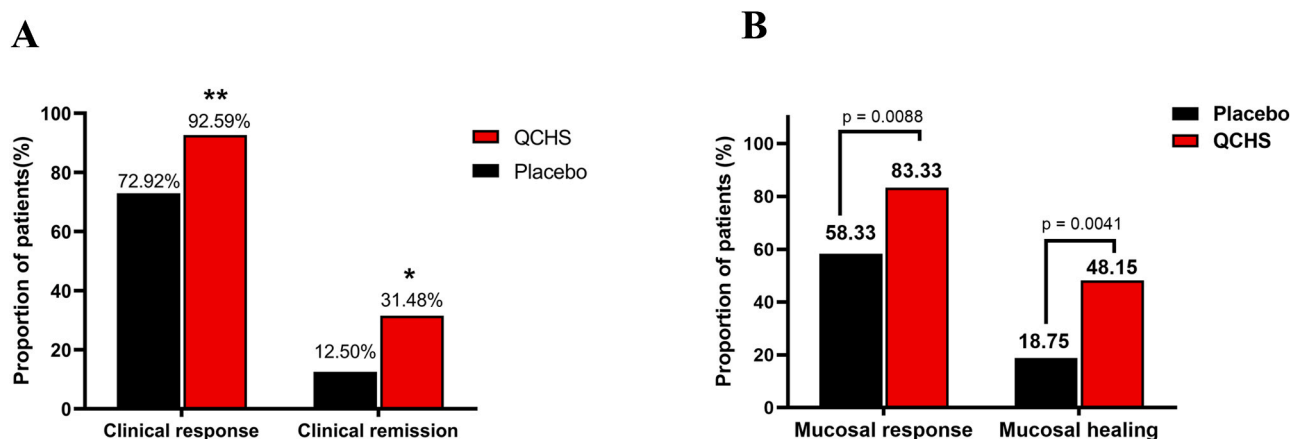


Fig. 2. Primary and secondary endpoints at week 12. A. Primary endpoints at the study end point of week 12. B. Secondary endpoints at the study end of week 12. The clinical response was defined as a 3-point decrease in the Mayo score and a decrease of at least 30% from baseline, along with a decrease of at least 1 point in the rectal bleeding subscore or an absolute rectal bleeding score of 0–1. Clinical remission was defined as a Mayo score ≤ 2 and no subscores with a value greater than 1. Mucosal response was defined as a decrease of at least 1 point in the mucosa subscore. Mucosal healing was defined as an absolute mucosa subscore of ≤ 1 .

week 12 was 18.75% (9/48) and 48.15% (26/54, $P = 0.0041$, Fig. 2B) in patients who received placebo and QCHS granules, respectively.

3.3. Other endpoints

There was a gradual increase in patients who received QCHS granules during the study period and achieved mucus/bloody stool remission (Fig. 3A), and a significantly greater percentage of patients receiving QCHS granules had a reduction in rectal bleeding as compared with patients receiving the placebo at 8 weeks (57.41% vs. 29.17%, $P = 0.0094$), 10 weeks (64.81% vs. 39.58%, $P = 0.0174$), and 12 weeks

(70.37% vs. 47.92%, $P = 0.0361$). Additionally, the stool frequency subscore was decreased during this study in both two groups (Fig. 3B). There was significantly different between patients achieved mucus/bloody stool remission in different group at 12 weeks (70.37% vs. 47.92%, $P = 0.0361$), as for stool frequency, it was not significantly different between patients who received QCHS granules and placebo (Fig. 3C).

3.4. Safety assessment

The most common AEs linked to herb medicine are abnormal liver

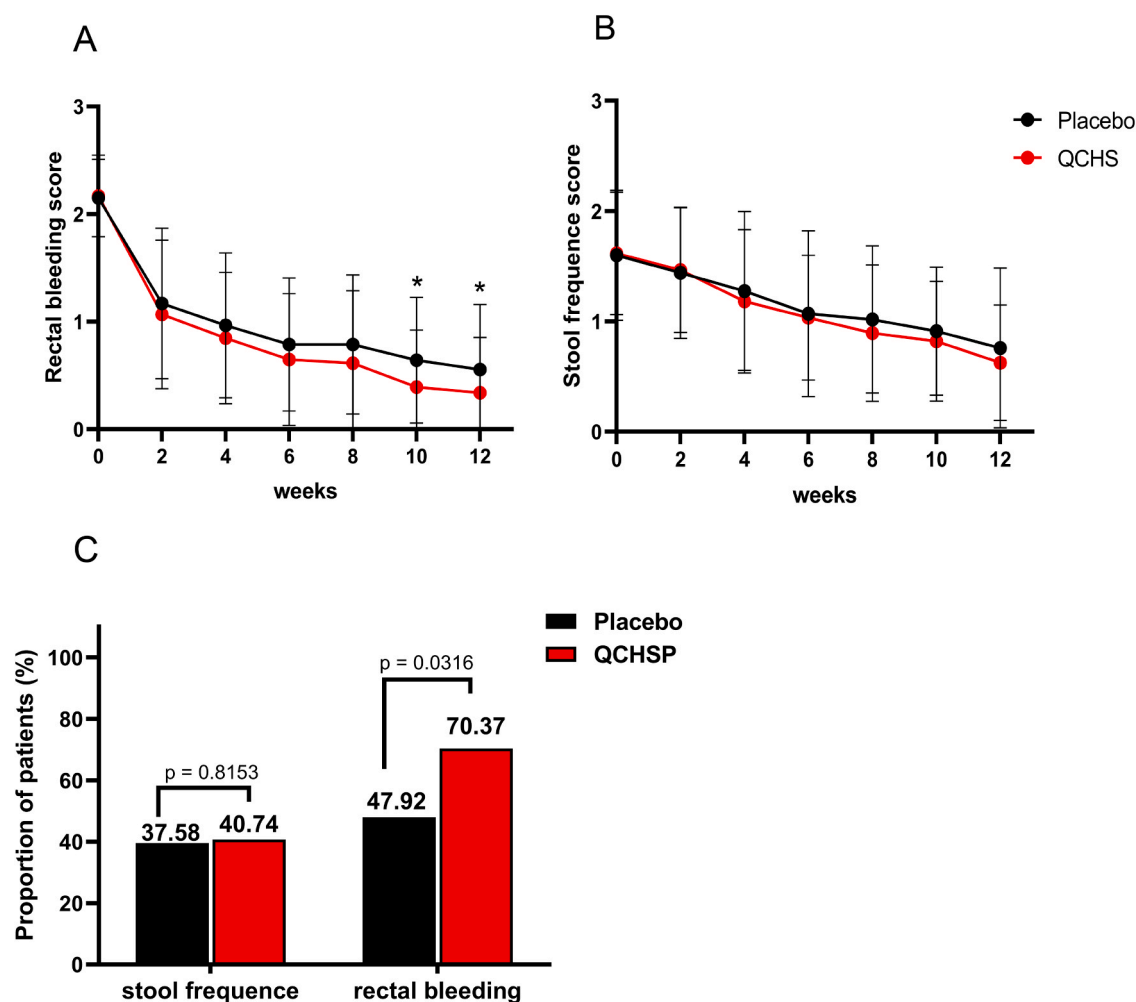


Fig. 3. Changes in other outcomes over time. A. Rectal bleeding subscore during the clinical trial. B. Stool frequency subscore during the clinical trial. C. Percentage of patients that achieved stool frequency remission and rectal bleeding remission at the study end of week 12. Asterisks indicate a significant difference between the treatment and placebo groups at each visit (Fisher's exact test, * $P < 0.05$).

and kidney function. In this study, All 119 randomized patients were included in the safety assessment (Table 3). Twenty-seven (27/60, 45%) AEs in the QCHS group were reported, and 27 (27/59, 45.76%) AEs in the placebo group were reported. Study discontinuation due to AEs occurred in 3.33% and 3.39% of patients receiving QCHS granules or placebo, respectively. The most common AEs, liver dysfunction and renal dysfunction, occurred with comparable frequency in patients. There were no significant differences in the incidence of overall AEs between the two groups during the study. The level of liver enzyme abnormalities was slit and the liver dysfunction of patients who received the placebo or QCHS granules was recovered spontaneously. At the endpoint, there were 3 patients with higher level of ALT and 1 patient with higher level of AST (Table 4).

4. Discussion

UC is a long-term disease that influences the quality of life in patients. Maintaining the remission stage and inhibiting the development of disease are vital in the management of UC. During last few decades, great progress has made in UC treatments, and monoclonal antibodies against TNF- α , fecal microbiota transplantation (FMT), and many other medicines have been used in clinical practice. However, the first-line treatment for mild to moderate active UC [37,38] is still 5-ASA. Patients with 5-ASA-refractory UC may opt to utilize corticosteroids [39], probiotics [40], monoclonal antibodies against TNF- α and humanized

monoclonal integrin blocker, which may lead to many other potential serious side effects for patients with UC and a higher economic burden [41]. Therefore, there is a need to develop new drugs that can induce remission in moderately active, 5-ASA-refractory UC, which would give another option for these patients.

Natural medicines have become the main resource of new drug development and about half of newly developed drugs are derived from herbs, either directly or indirectly [42]. Thus, the use of natural medicines for the treatment of UC has increased over the years [18,20,22,24]. Although clinical trials and experimental studies have demonstrated that various herb-derived bioactive compounds, such as curcumin [20] and *Andrographis paniculata* extract [22], may be potential candidates for UC treatment, few of these treatments perform satisfactorily as compared with conventional treatments. The pathogenesis of UC is thought to be system-level abnormalities rather than defects in the immune system [43]. Therefore, the traditional style, "one-target-one-drug", of drug development has been extended to drug-network interactions [44], and the therapeutic principle of TCM, "multiple compounds against multiple targets" [45], better correlates to this new drug development model.

Based on the traditional Chinese medicine theory and combined with clinical practice, our group created the QCHS formula, which is a complex system containing 11 herbs [46]. Previously, our group reported that QCHS granules alleviated chemical-induced colitis in rats by preventing inflammatory responses and suppressing apoptosis through

Table 3
Summary of adverse events and safety population.

Adverse events no. (%)	QCHS granule (n = 60)	Placebo (n = 59)	P value
Total AEs	66	60	
Total patients with AEs	27 (45.00)	27 (45.76)	0.933409
Related to study drug	23 (38.33)	15 (25.42)	0.130975
Leading to study drug Discontinuation	2 (3.33)	2 (3.39)	1.000000
Serious AEs	0	0	
Liver dysfunction			
ALT	11 (18.33)	12 (20.34)	0.781755
AST	6 (10.00)	5 (8.47)	0.776194
AKP	3 (5.00)	3 (5.08)	1.000000
TBIL	5 (8.33)	9 (15.25)	0.241358
GGT	4 (6.67)	4 (6.78)	1.000000
Renal dysfunction			
BUN	0 (0.00)	2 (3.39)	0.243697
SCR	0 (0.00)	1 (1.69)	0.495798
Electrocardiographic abnormality	0	0	
Other abnormal laboratory test results			
urine leukocyte	8 (13.56)	2 (3.39)	0.104297
urine erythrocyte	5 (8.47)	5 (8.47)	1.000000
urine protein	0 (0.00)	3 (5.08)	0.118724

AEs, adverse effects; ALT, alanine aminotransferase; AST, aspartate amino-transferase; AKP, alkaline phosphatase; TBIL, total bilirubin; GGT, γ -glutamyl transpeptidase; BUN, urea nitrogen; SCR, serum creatinine concentration.

Table 4
Summary of adverse events of liver enzyme abnormalities at each visit.

		QCHS granule (n = 60)	Placebo (n = 59)	P value
ALT	week 2	4 (6.67)	5 (8.33)	0.743012
	week 4	1 (1.67)	4 (6.67)	0.206833
	week 8	3 (5.00)	3 (5.00)	1.000000
	week 12	3 (5.00)	0 (0.00)	0.243697
AST	week 2	2 (3.33)	2 (3.33)	1.000000
	week 4	1 (1.67)	2 (3.33)	0.618617
	week 8	2 (3.33)	1 (1.67)	1.000000
	week 12	1 (1.67)	0 (0.00)	1.000000

the MEK/ERK signaling pathway [46]. Various bioactive compounds derived from herbs in the QCHS formula have shown therapeutic effects in animal models of colitis through different ways. Wogonoside [47], derived from Huangqin, demonstrated anti-inflammatory effects via inhibition of the NLRP3 inflammasome. Baicalin [48], another active compound from Huangqin, restored gut microbiota-metabolism homeostasis and regulated the balance of Treg/Th17 cells. IMP [49], one of the most abundant coumarins in Baizhi, served as a PXR agonist to attenuate DSS-induced colitis by suppressing the NF- κ B-mediated proinflammatory response. Berberine hydrochloride [50] has also shown anti-inflammatory effects by blocking the IL-6/STAT3/NF- κ B signaling pathway. However, a lack of evidence from clinical trials has prevented QCHS granules, a type of empirical medicine, from being recommended to patients with UC.

Our study was the first to perform a randomized, double-blind, placebo-controlled study to evaluate the therapeutic effect of QCHS granules on UC. All participants were patients with moderate UC, who still have bloody stool four to six times a day after receiving mesalazine treatment. We found that the QCHS granules were superior to the placebo in inducing clinical efficacy (92.59% vs. 72.92%), clinical remission (31.48% vs. 12.5%), mucosal response (83.33% vs. 53.33%), and mucosal healing (48.15% vs. 18.75%). As shown in our study, the presence of mucosal healing was higher than the presence of clinical remission.

Although only 31.48% of the patients receiving QCHS granules achieved clinical remission, a subjective primary endpoint based on Mayo score, this was significantly higher than the placebo group, where only 12.5% of the patients achieved clinical remission. In the past few years, clinical remission is an important endpoint for patients with UC [51]. However, there are two components, rectal bleeding and stool frequency, that are dependent on patients' subjective perceptions, which may lead to false-positives in a study. Increasing guidelines have suggested that the goals of UC treatment include both the induction of clinical remission and mucosal healing, which is the absence of mucosal ulcerations and erosions [1,52]. Because of this, we also selected mucosal healing, a component of the Mayo score, as a secondary endpoint of this study. Our current study showed that QCHS granules improve mucosal healing, with a significantly greater percentage of patients in the QCHS granules group achieving mucosal healing after 12 weeks as compared with the patients in the placebo group (48.15% vs. 18.75%, respectively). Interestingly, our results also showed that the percentage of patients achieving mucosal healing was greater than the percentage of patients in clinical remission in both groups. As for other endpoints, over half of the patients (70.37%) receiving QCHS granules reported a decrease in the level of pus-blood stool, which was significantly higher than patients in the placebo group. However, QCHS granules were not superior to the placebo in relieving abdominal pain and diarrhea in the patients.

Our study has some limitations. First, the clinical response rate of placebo-treated was too high, which may be due to some subjective outcomes of the Mayo score. Compared with the patient-reported outcomes and physician-assessment outcomes, evaluators tended to score as improved may be the main reason for too high clinical response rate. Second, previous study has shown that central review of endoscopic images is critical to the conduct of induction studies in UC [53]. In this study, endoscopists were from multiple centers, and as such, the lack of an independent central reader may have affected endoscopy outcomes. Whatmore, several studies have shown that histological remission, a target distinct from endoscopic mucosal healing, can be used as a clinical endpoint [54] and predicts lower rates of corticosteroid use and acute severe colitis in patients with UC [55]. Due to the lack of a standardized system for how histological endpoints are scored, reported, and defined [56], histological remission was not evaluated as a clinical endpoint in this study. Last but not least, the lesion locations were different in participants, while most can be induced as left hemicolitis or extensive colitis. And there were no-correlation between any particular subtypes patients and therapy respond. Patients with proctitis were included, which is not usually the case in UC trials. As these results, we will do more research in proctitis.

5. Conclusion

QCHS granules are efficacious in introducing clinical remission and mucosal healing, as well as in relieving mucus/blood stool in patients with moderately active and 5-ASA-refractory UC.

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Author Contributions

Guarantor of the article: Lei Zhu. Hong Shen and Lei Zhu contributed to the conception of the study, interpretation of the data, and critical revision of the manuscript. Shengsheng Zhang, Wenxia Zhao, Shunping Ren, Xiao Ke, Qinghua Gu, Zhipeng Tang, Jingri Xie, Suning Chen, Yan Chen, Lu Zhang, Kai Zheng, Yajun Liu, Peiqing Gu and Jiafei Chen contributed to conduction of the study. Jiandong Zou and Zhaofeng

Shen contributed to the analysis and interpretation of the data. Jingyi Hu drafted the manuscript. All authors approved the final version of the manuscript for submission.

Conflict of interest statement

The authors declare that they have no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.biopha.2021.111580](https://doi.org/10.1016/j.biopha.2021.111580).

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